

FARM MANAGEMENT SYSTEM

STUDENT NAME:P.SAKTHI (Leader)

YEAR: III

COLLEGE CODE:304

COLLEGE NAME: ARIGNAR ANNA GOVT.ARTS COLLEGE FOR WOMEN,
WALAJAPET.

DEPARTMENT: COMPUTER SCIENCE

SEMESTER: V

TEAM MEMBERS : J. PAPITHA
R. THRISHA
D.AKALYA
R. KEERTHANA

User Story:

As a Farmer, Farm Manager, Buyer, and System Administrator,
I want a centralized Salesforce-based Farm Management System to efficiently record, organize, and manage information related to farms, crops, farmers, and buyers,
So that agricultural information can be maintained in a structured and easily accessible manner, reducing the dependency on manual records and improving the accuracy of data.
The system enables users to update farm details, crop production information, farmer details, and buyer transactions through a centralized platform. Data validation helps maintain accurate and complete records, while reports and dashboards provide a clear view of agricultural operations and support effective monitoring and analysis.
The system also provides role-based access and secure cloud-based data storage, allowing authorized users to access the information relevant to their responsibilities. Overall, the Farm Management System helps reduce paperwork and manual errors, improves data accessibility and transparency, and makes the management of agricultural operations more organized and efficient.

1. INTRODUCTION

1.1 Project Overview:

The Farm Management System is a Salesforce-based application designed to digitize and streamline the management of agricultural operations. The system provides a centralized platform for maintaining important information related to farms, crops, farmers, and buyers.

Traditional agricultural record management may involve maintaining information across manual registers and separate files, which can make data management difficult and time-

consuming. The proposed system organizes this information within a centralized Salesforce environment, making records easier to maintain, access, and update.

The system supports structured data entry and validation to improve the accuracy and completeness of records. Relationships between different records help users maintain connected information, while reports and dashboards provide a clear view of agricultural activities and support monitoring and analysis.

By bringing essential farm-related information into a single platform, the Farm Management System aims to reduce paperwork, minimize manual errors, improve accessibility of information, and make agricultural operations more organized and efficient.

1.2 Purpose:

The main purpose of the Farm Management System is to provide a centralized and organized platform for managing agricultural information using Salesforce. The system is designed to simplify the process of recording, updating, accessing, and monitoring important farm-related data.

The system aims to:

- Digitize farm, crop, farmer, and buyer information.
- Reduce dependency on manual registers and paperwork.
- Improve the accuracy and consistency of stored information.
- Provide easy and organized access to agricultural records.
- Support real-time monitoring through reports and dashboards.
- Improve transparency in farm operations and related transactions.
- Help users analyze available information and make informed operational decisions.

- Increase the overall efficiency of agricultural data management.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain accurate records of farm activities and crop details.	Managing records manually is difficult and time-consuming.	Information is stored in notebooks or separate files.	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and manage daily activities efficiently.	It is difficult to obtain updated information quickly.	Data is stored in spreadsheets and separate files.	Stressed and inefficient while preparing reports.
PS-3	Buyer	Find reliable information about crop availability and purchasing details.	Updated information is not always available.	There is no centralized platform for accessing crop information.	Uncertain and dissatisfied.
PS-4	System Admin	Manage user access and protect system information securely.	Managing permissions manually is difficult.	Different users require different access levels.	Concerned about unauthorized access and data security.

Problem Statements – Farm Management System

	I AM (Customer)	I'M TRYING TO	BUT	BECAUSE	WHICH MAKES ME FEEL
 FARMER	 I want to maintain accurate farm and crop records.	 Maintain farm activities and crop details properly.	 Managing records manually is difficult and time-consuming.	 Information is stored in notebooks or separate files.	 Frustrated and confused.
 FARM MANAGER	 I want to monitor productivity and manage farm activities efficiently.	 Collect information and prepare productivity reports.	 Difficulty obtaining accurate and updated information quickly.	 Spreadsheets, files and delayed reports cause inaccuracies.	 Stressed and inefficient.
 BUYER	 I want reliable information about crop availability and purchasing details.	 Find crops and make informed purchasing decisions.	 Updated information is not always available.	 There is no centralized platform for accessing crop information.	 Uncertain and dissatisfied.
 SYSTEM ADMINISTRATOR	 I want to maintain system security and manage user access.	 Create users and manage permissions effectively.	 Difficulty controlling access and protecting information.	 Different users require different access levels.	 Concerned and insecure.

2.2 Empathy Map Canvas

Target Users:

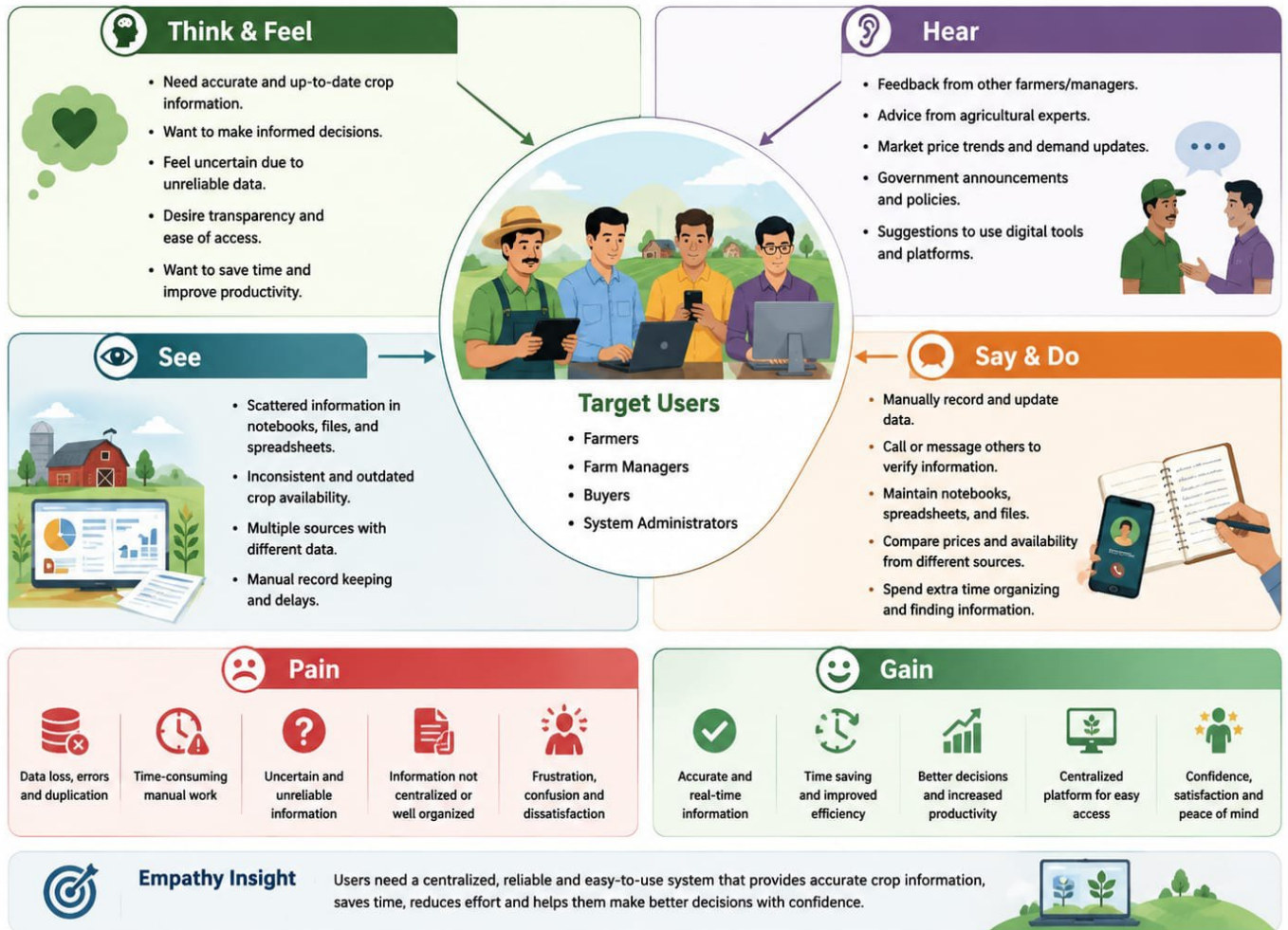
- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- Think & Feel: What the user thinks and feels.
- Hear: What the user hears from others.
- See: What the user observes around them
- Say & Do: What the user says and does.
- Pain: Problems and difficulties faced by the user.
- Gain: Benefits and improvements expected by the use

EMPATHY MAP – FARM MANAGEMENT SYSTEM

Understanding our users to build a better agricultural ecosystem



2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

1. Manual Farm Records Maintaining farm and crop details using notebooks and paper registers.

This method is simple but time-consuming and may lead to errors or data loss.

2. Excel-Based Farm Management Using Excel spreadsheets to store and manage farm,

crop, farmer, and buyer information. It improves organization but may become difficult when data increases.

3. Web-Based Farm Management System Developing a web application to manage agricultural activities and access information through the internet.
4. Mobile Farm Management Application Creating a mobile application that allows farmers and managers to update and access farm information using smartphones.
5. Salesforce-Based Farm Management System (Selected Idea) Developing a centralized Salesforce-based system to manage farm, crop, farmer, and buyer information using custom objects, automation, reports, and dashboards.

Selected Idea

Salesforce-based Farm Management System.

Reason for Selection:

It provides centralized data storage, improved data accuracy, secure access, automation, and real-time reports and dashboards.



6. REQUIREMENT ANALYSIS PHASE

6.1 Customer Journey Map

Stages:

1. Login

Users log in to the Salesforce application using their credentials.

2. Access Farm Management App

Users open the Farm Management System through the Salesforce App Launcher.

3. Enter Farm and Crop Data

Users add and update farm details, crop information, and production data.

4. Register Farmers and Buyers

Farmer and buyer details are entered and maintained in the system.

5. Generate Reports

Reports are generated using the stored farm, crop, farmer, and buyer data.

6. Analyze Results

Users analyze productivity, sales, and other agricultural information.

7. Decision Making

Users make informed decisions based on reports and dashboard insights.



6.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication The system should allow users to log in securely using valid credentials.
FR-2	Farm, Crop, Farmer, Buyer Management The system should allow users to manage farm, crops, farmers, and buyer records.
FR-3	Data Entry and Validation The system should validate entered information and prevent incorrect or incomplete data.
FR-4	Relationship Mapping The system should establish relationships between farms, crops, farmers, and buyers.
FR-5	Report Generation The system should generate reports related to farm activities, crop production, and buyer transactions

FR-6	Dashboard Visualization The system should display important information through dashboards and charts.
FR-7	User Role Management The system should provide role-based access according to user responsibilities.

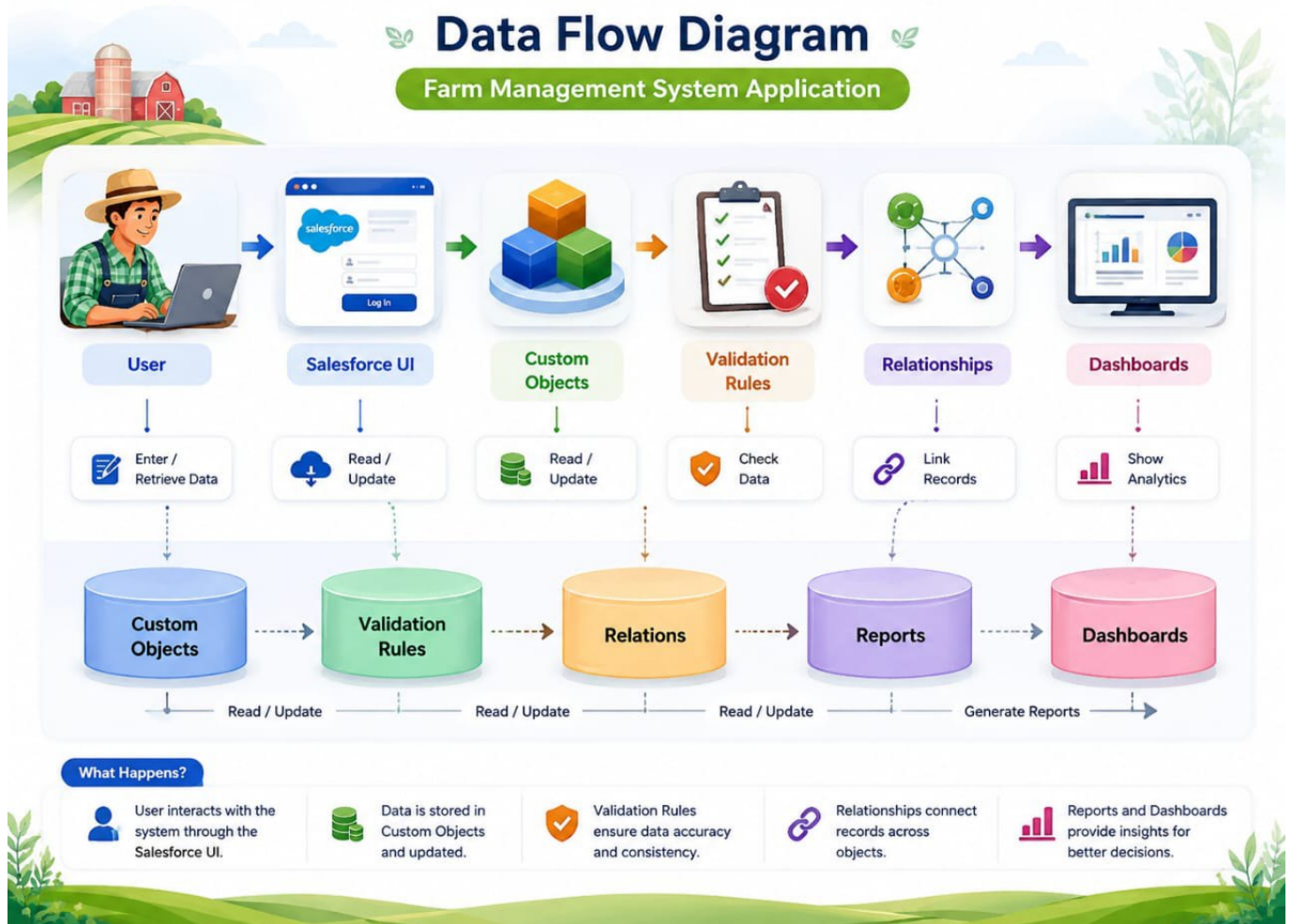
Non-Functional Requirements

NFR	Description	Example
Usability	The system should provide a simple and user-friendly Salesforce Lightning interface.	Users can easily navigate through the Farm Management System using the Salesforce Lightning interface.
Security	The system should protect data using profiles, roles, and sharing rules.	Only authorized users can access farm and crop information based on their assigned roles.
Reliability	The system should store data securely	The system should save records correctly without losing important information.
Performance	Reports and dashboards should load quickly and respond efficiently.	Reports and dashboards should load quickly when users request information.
Availability	24/7 access to the system should be accessible through Salesforce cloud services whenever required.	Users should be able to access the system through the internet whenever required.
Scalability	The system should support an increasing number of farms, users, and records.	The system should support more farms, users, and records in the future.

6.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



6.4 Technology Stack

Layer	Technology
Presentation	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce custom objects and Cloud database
Automation	Salesforce flows
Security	Controls user access and protects system information

7. PROJECT DESIGN PHASE

7.1 Problem–Solution Fit

The Farm Management System identifies common agricultural management problems and provides suitable digital solutions. It replaces manual record keeping with a centralized Salesforce platform to improve accuracy, efficiency, and decision-making.



7.2 Proposed Solution

Functional Requirements

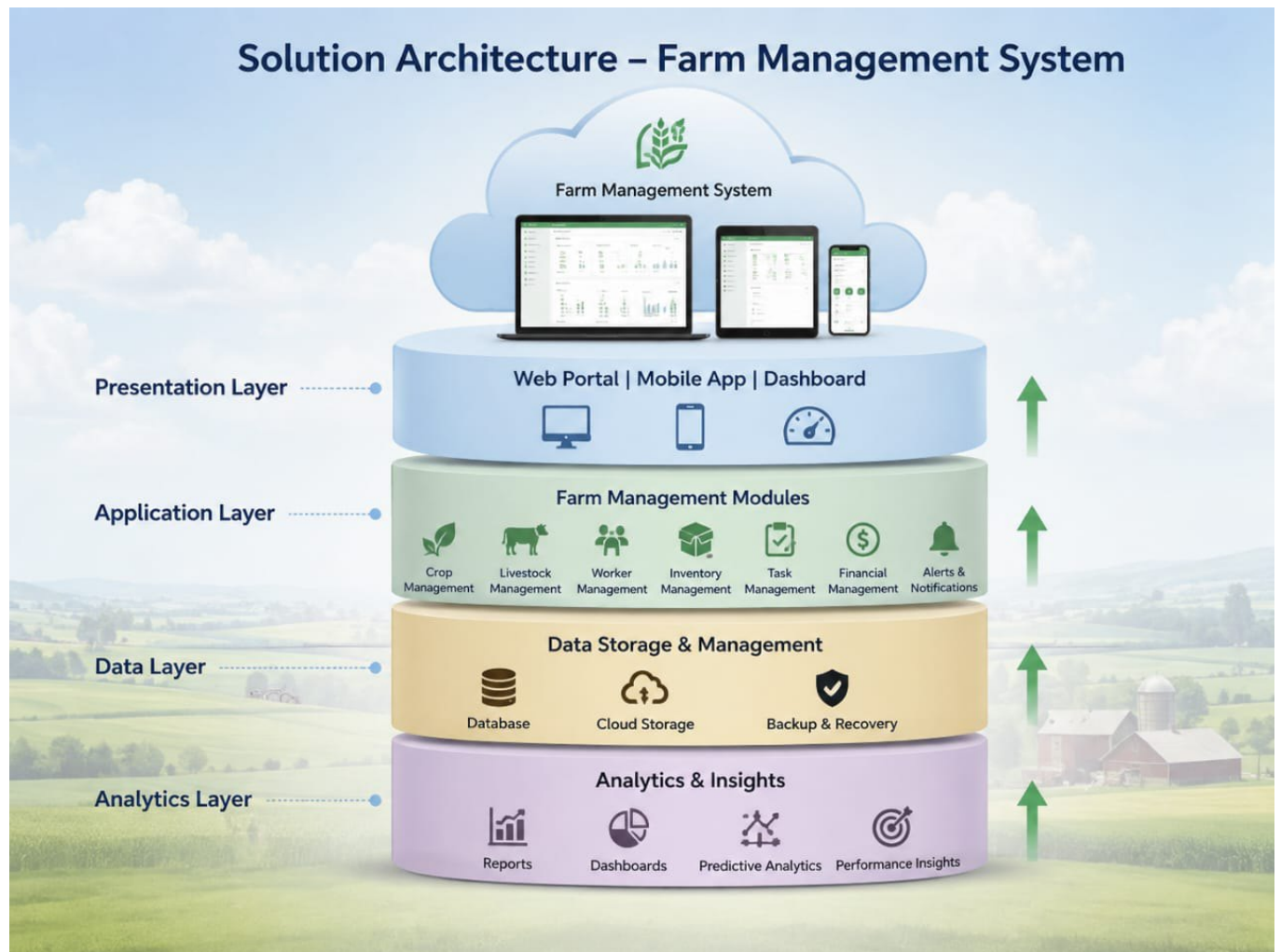
Parameter	Description
Problem	Farmers and farm managers face difficulties in maintaining manual records, tracking crop details, and accessing updated information.

Solution: A centralized Salesforce-based system manages farm, crop, farmer, and buyer information digitally.
Innovation: Uses custom objects, validation rules, automation, reports, and dashboards to simplify farm management.
Impact: Reduces paperwork and errors while improving productivity, monitoring, and decision-making.
Business Model: Provides digital farm management services for farmers, farm managers, and agricultural organizations.
Scalability: Can support additional farms, users, records, and agricultural activities as the system grows.

7.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI provides an easy-to-use interface for users.
2. Application Layer – Validation Rules and Relationships manage application logic and data connections.
3. Data Layer – Salesforce Custom Objects store farm, crop, farmer, and buyer information.
4. Analytics Layer – Reports and Dashboards display useful information for monitoring and decision-making.
- 5.



8. PROJECT PLANNING & SCHEDULING

8.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	user Story / Task	Story Points	Priority	Team Members
Sprint-1	User authentication	USN-1	. Configure Salesforce user access.	3	High	Member1
Sprint-2	Record management	USN-2	Create Farm, Crop, Farmer, and Buyer objects.	5	High	Member2
Sprint-3	Data entry and validation	USN-3	Create fields and validation rules.	5	High	Member3
Sprint-4	Relationship mapping	USN-4	Establish relationships between objects	3	High	Member4
Sprint-5	User role management	USN-5	Configure profiles, roles, and sharing rules.	5	High	Member5
Sprint-6	Report generation	USN-6	Create reports for farm-related data.	3	High	Member1
Sprint-7	Dashboard visualization	USN-7	Create dashboards for data analysis.	4	Medium	Member2
Sprint8	Testing	USN-8	Test system functionality and review results.	4	High	All members

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	5	3 Days	19 sept 2026	19 sept 2026		
Sprint-2	5	3 Days	19 sept 2026	19 sept 2026		
Sprint-3	5	3 Days	19 sept 2026	20 sept 2026		
Sprint-4	5	3 Days	20 sept 2026	20 sept 2026		
Sprint5	5	3 days	20 sept 2026	21 sept 2026		
Sprint6	3	3 days	21 sept 2026	21 sept 2026		
Sprint7	4	3 days	21 sept 2026	21 sept 2026		
Sprint8	4	3 days	21 sept 2026	21 sept 2026	34	21/09/2026

9. Project Development Phase:

Project Description:

The Development Phase focuses on implementing the proposed Salesforce-Based Farm Management System according to the functional and non-functional requirements identified during the requirement analysis stage.

During this phase, the Salesforce Developer Org is configured and customized to manage farm-related information efficiently. Custom objects such as Farm, Crop, Farmer, and Buyer are created with the required fields and relationships.

Validation rules are implemented to maintain data accuracy and reduce incorrect entries. Profiles, roles, and sharing rules are configured to provide secure access to users based on their responsibilities. Salesforce Reports and Dashboards are also developed to display useful farm-related information and support decision-making.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is a cloud-based customer success platform designed to help organizations manage, analyze, and connect with their customers and business operations. In this project, Salesforce is used as the main platform to develop the Farm Management System. It helps manage farm-related data such as Farm, Crop, Farmer, and Buyer records through custom objects, validation rules, relationships, reports, and dashboards.

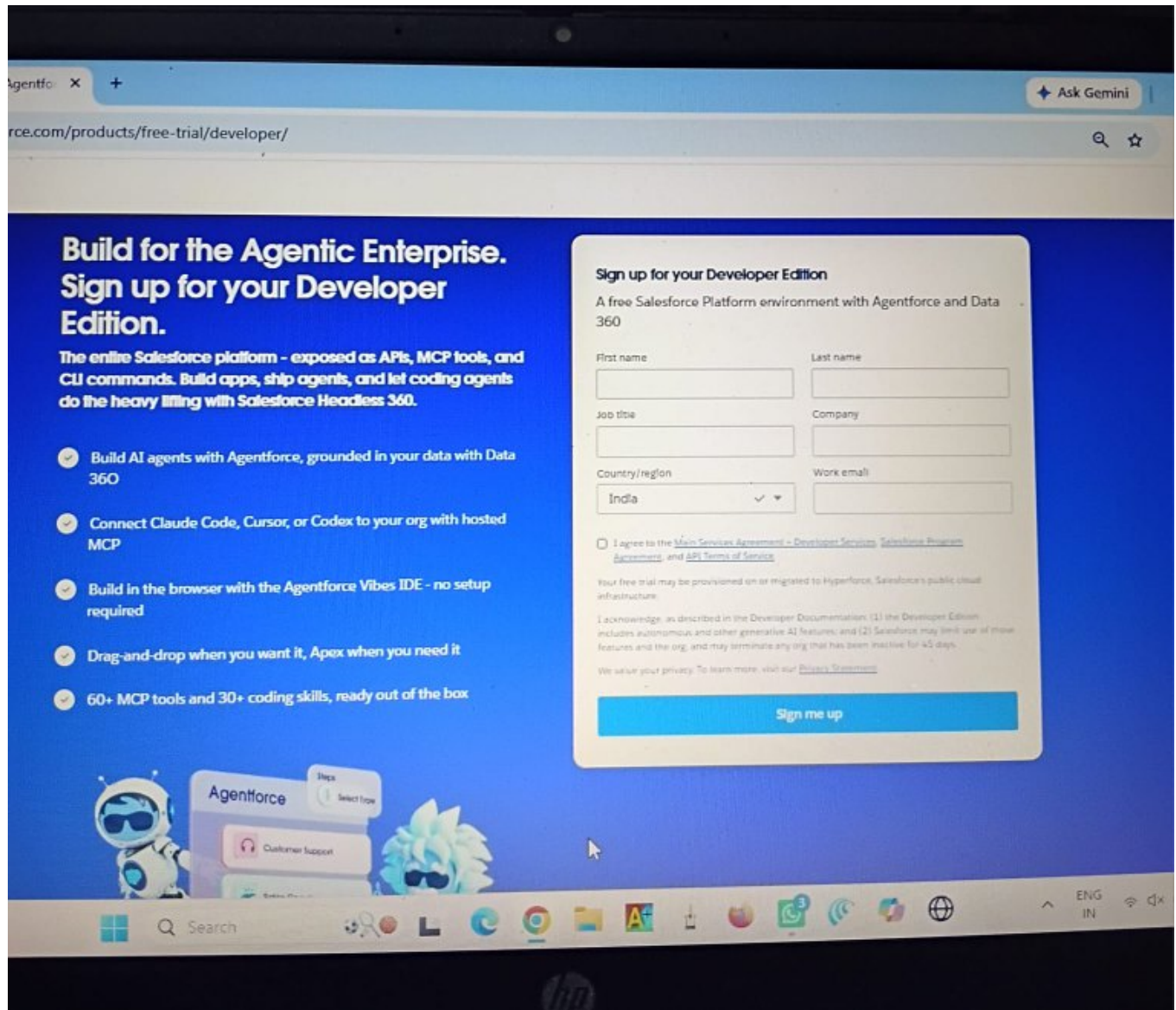
Salesforce provides various standard products and features that allow users to manage relationships, collaborate with employees and partners, store data securely in the cloud, and access business information from anywhere.

Activity1:

Creating a Salesforce Developer Org:

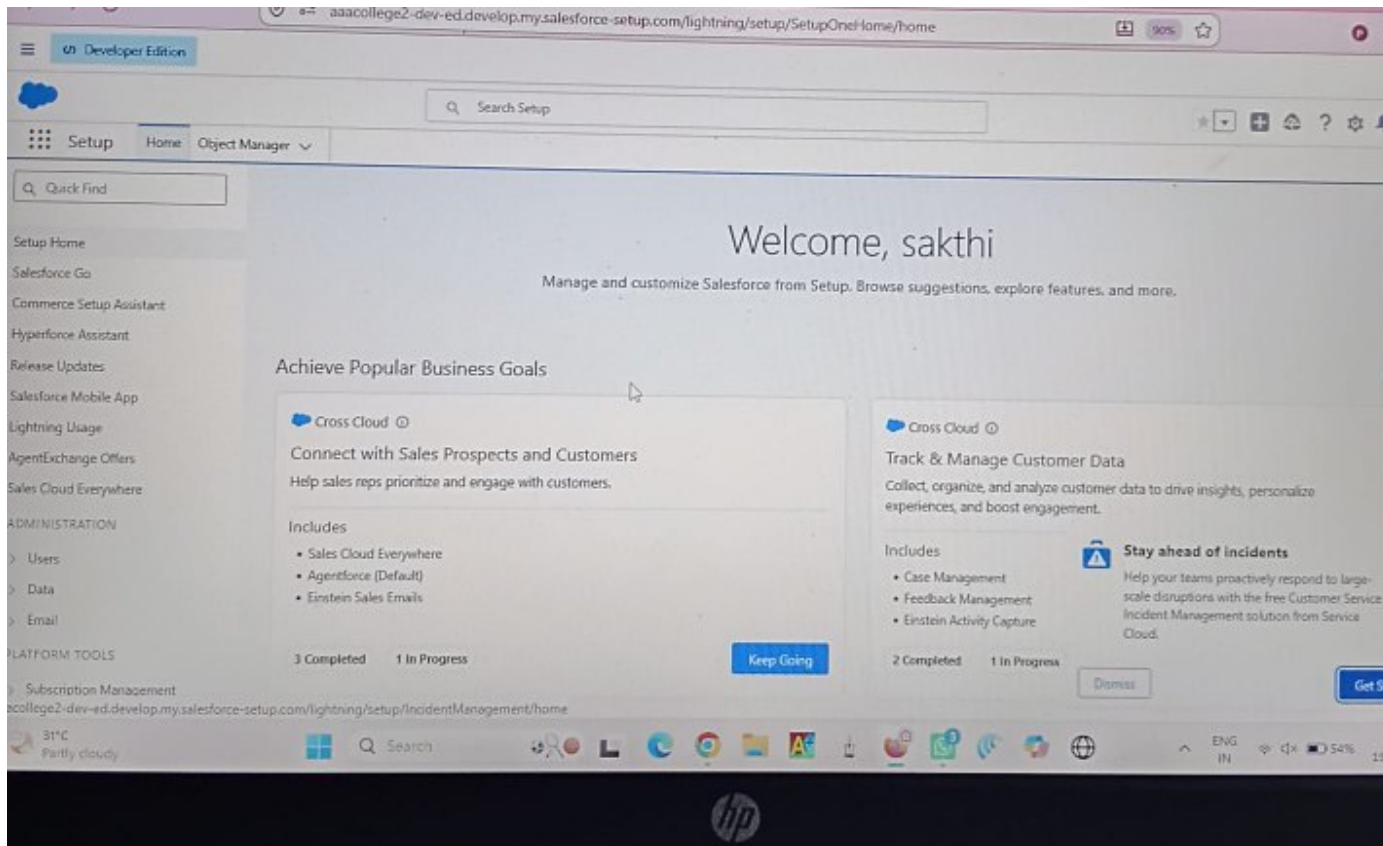
A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click Save.
5. Search login.salesforce.com



6. By using username and password you can into the Salesforce org. The homepage will appear below

7.



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that allow you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

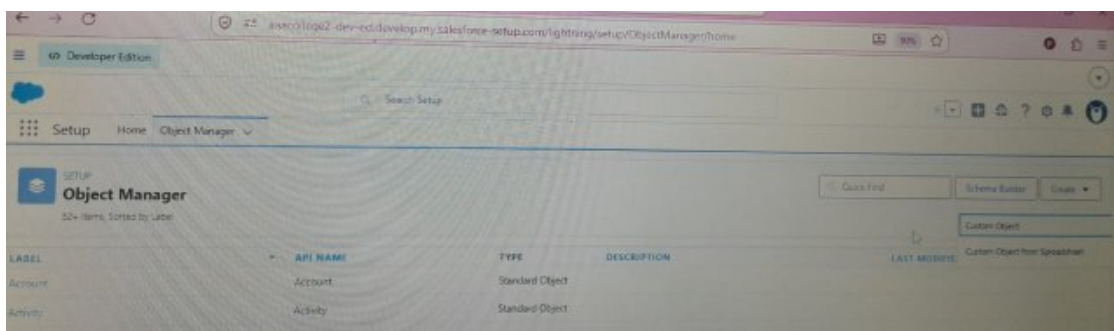
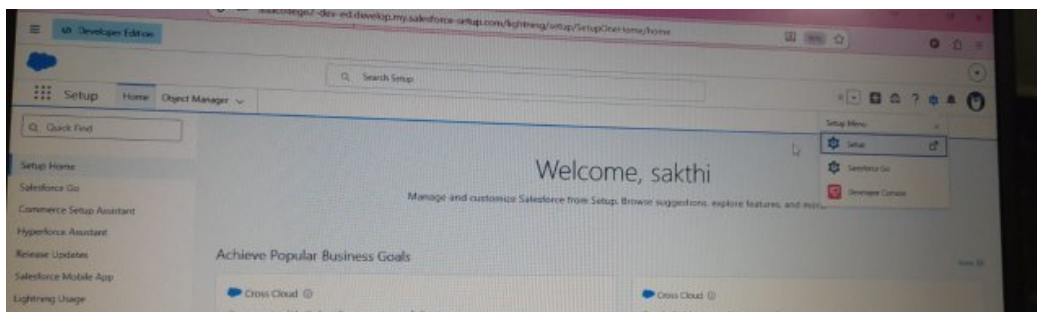
1. Farm
2. Crop
3. Farmer
4. Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

Date type: **Text** Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing (...)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status

☐ In Development

☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more](#)

- ☒ Allow Search

Object Creation Options (Available only when custom object is first created)

- ☐ Add Notes and Attachments related list to default page layout
- ☐ Launch New Custom Tab Wizard after saving this custom object

[Save](#) [Save & New](#) [Cancel](#)

7. In the Optional Features section, select Allow Reports and Track Field History.
 8. In the Deployment Status section, ensure Deployed is selected.
 9. In the Search Status section, select Allow Search.
 10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
-
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look at the extreme right, you will find a Create Dropdown, click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look at the extreme right, you will find a Create Dropdown, click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

- 1. Click on the gear icon and then select Setup.
- 2. Click on the object manager tab just beside the home tab.
- 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Milestone 3

- Tab: Tab

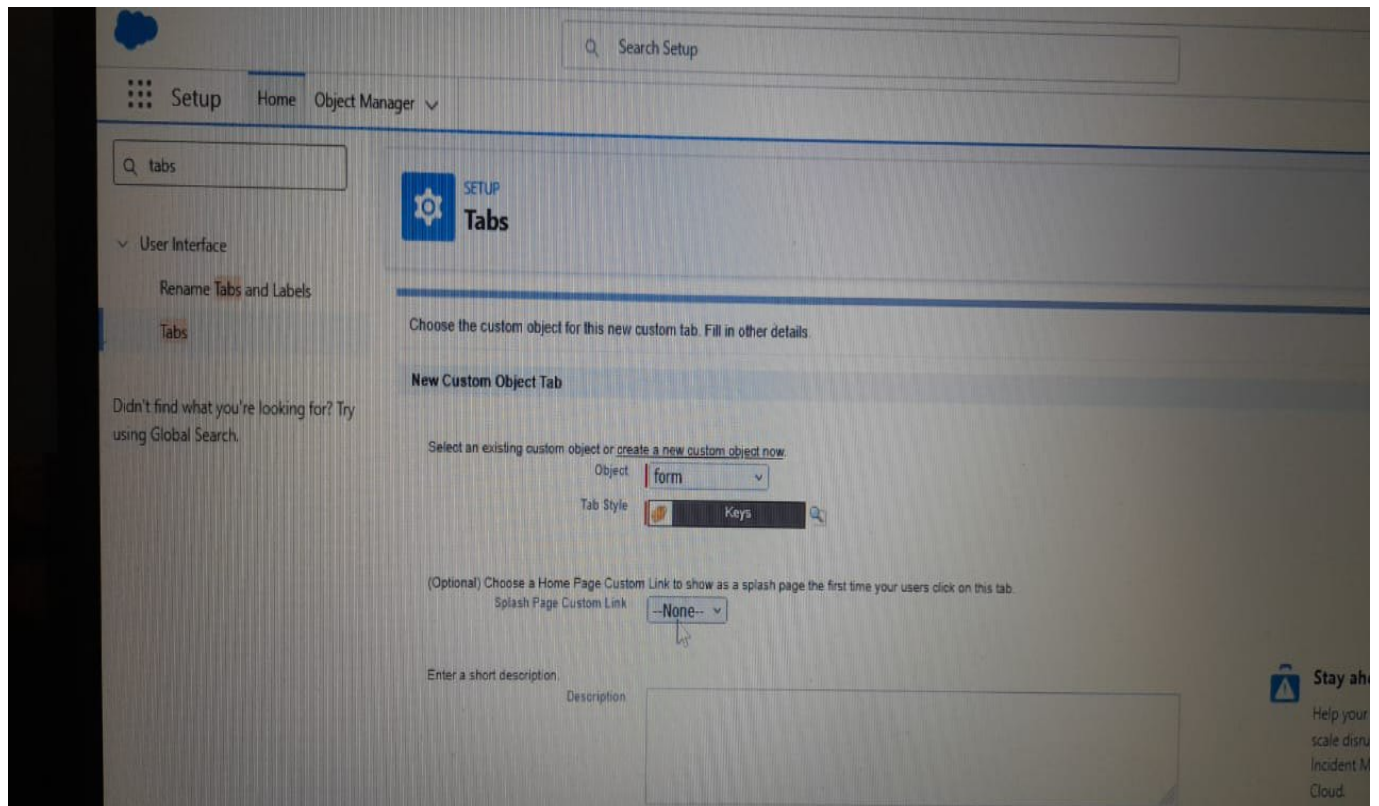
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

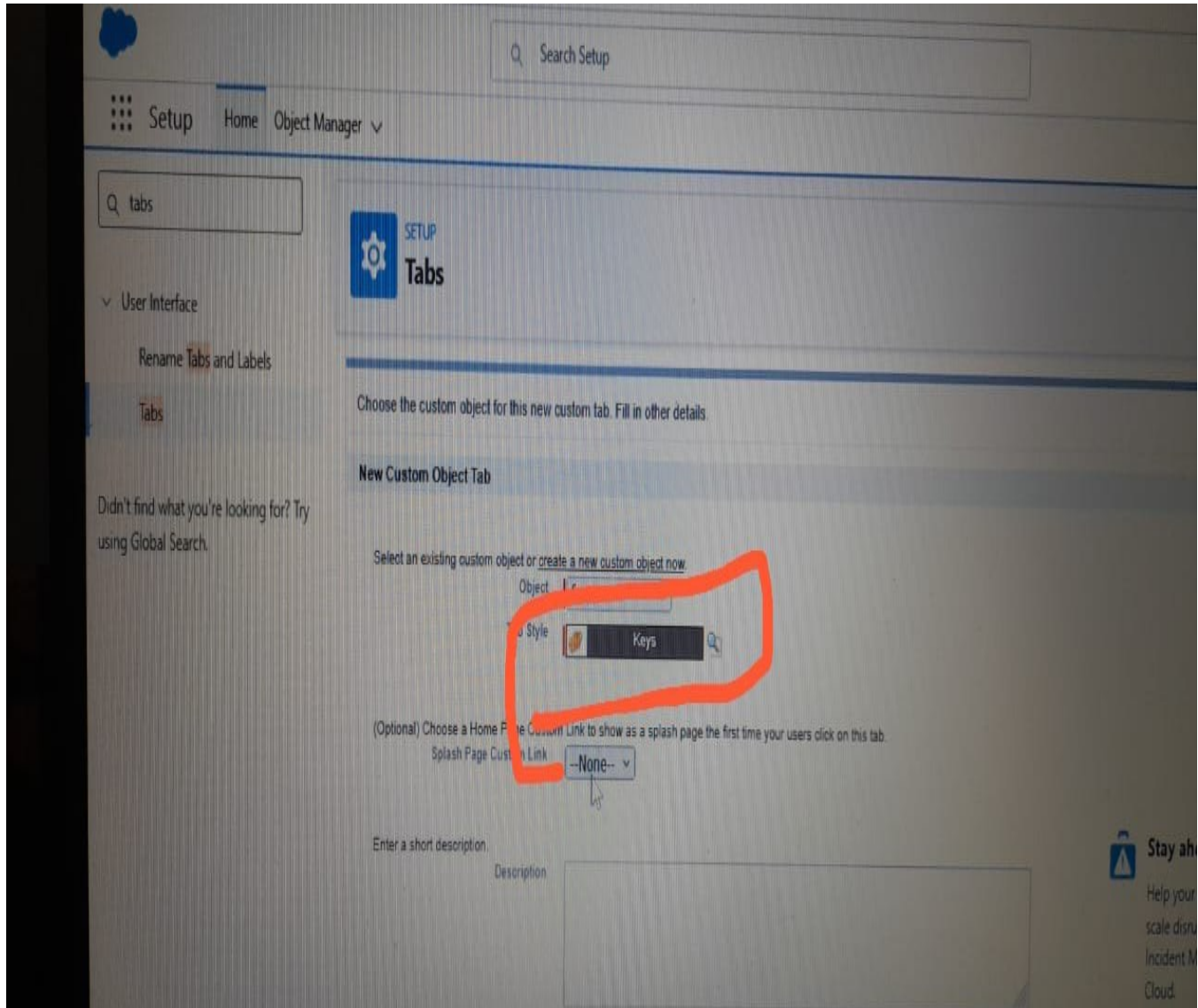
Now create a custom tab.

- 1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.

3.



4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

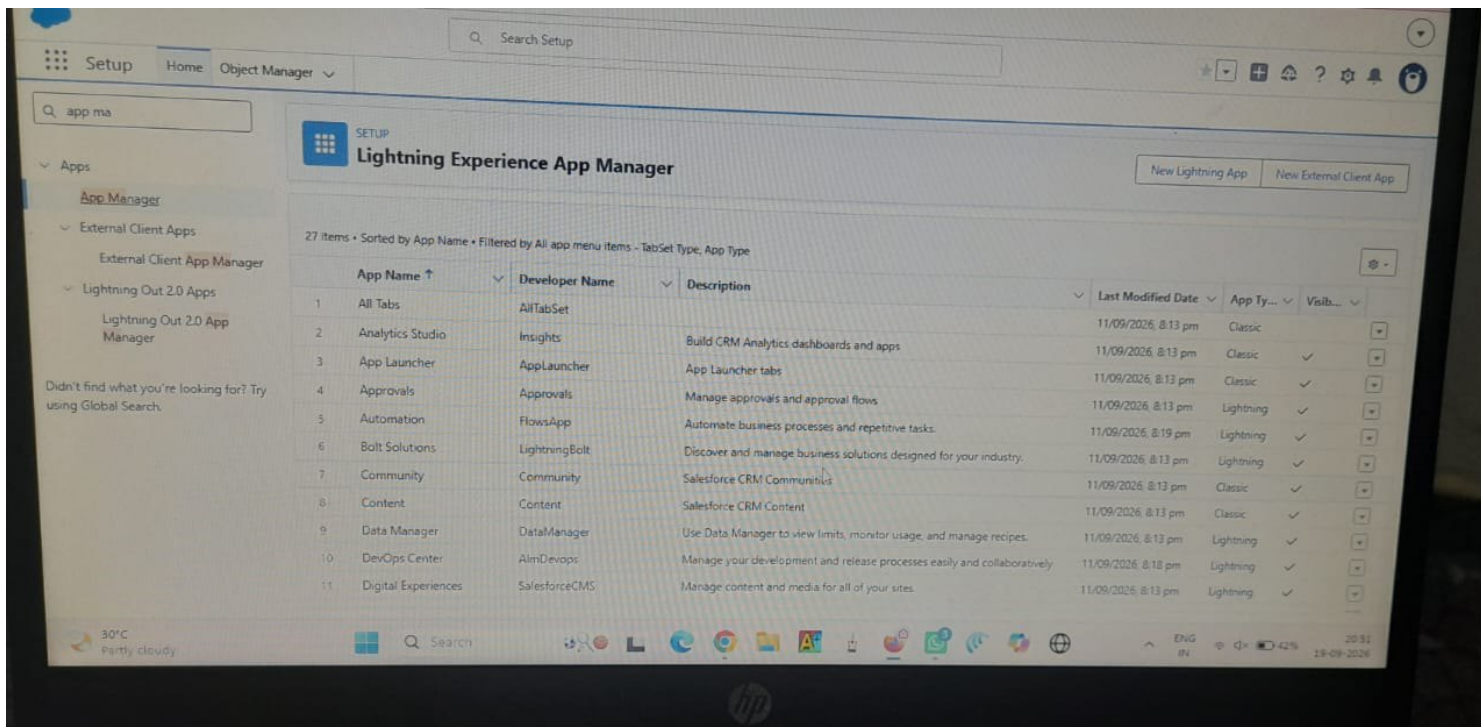
What is an App?

A Salesforce app is a collection of tabs, objects, and features that are grouped together to help users perform specific business tasks efficiently.

An app provides an organized interface where users can access related records and functionalities from a single place.

In this project, the Farm Management System App is created to provide easy access to the Farm, Crop, Farmer, and Buyer objects. It helps users navigate between different modules and manage agricultural information efficiently. Activity 1:

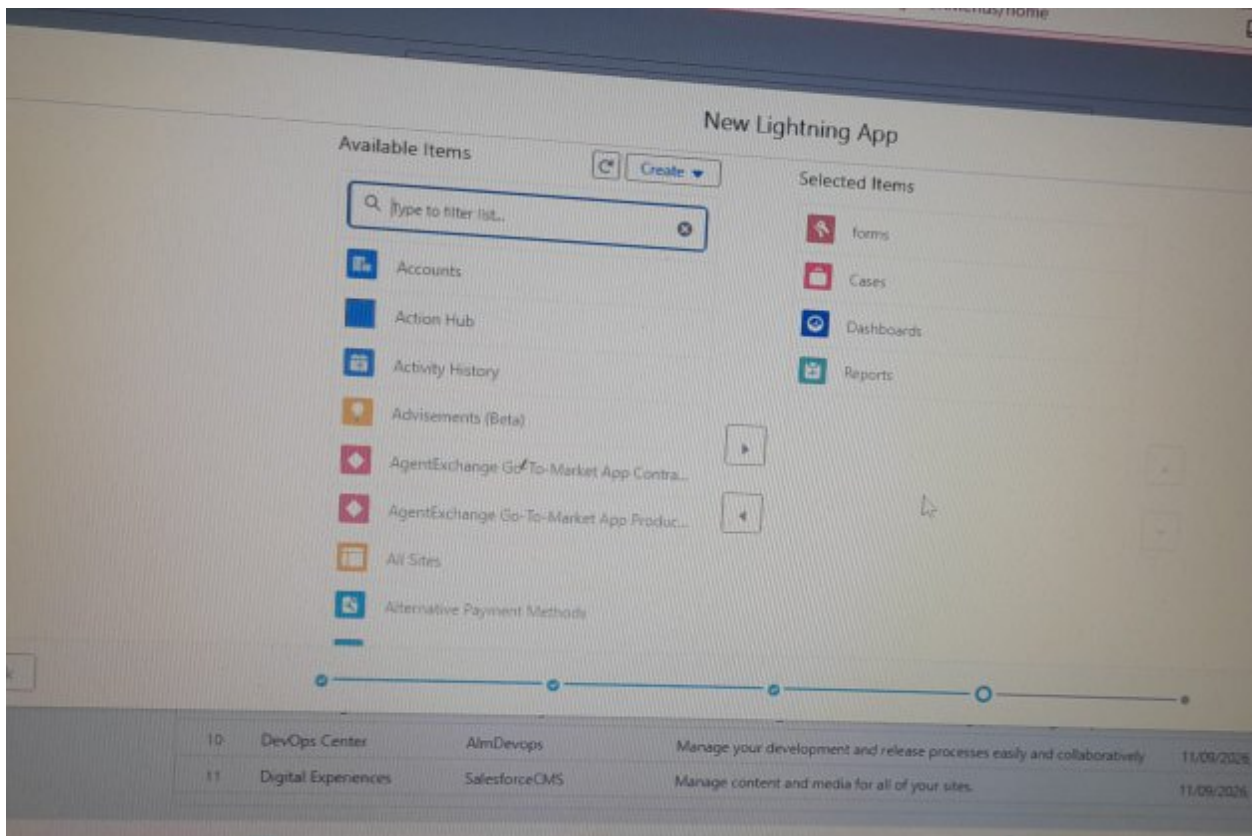
Create the Farm Management System app



- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next. From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.

10	DevOps Center	AlmDevops	Manage your development and release processes easily and collaboratively	11/09/2026, 8:18 pm	Lightning	✓
11	Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	11/09/2026, 8:13 pm	Lightning	✓

3. Navigation menu-Displays the tabs available inside the app.



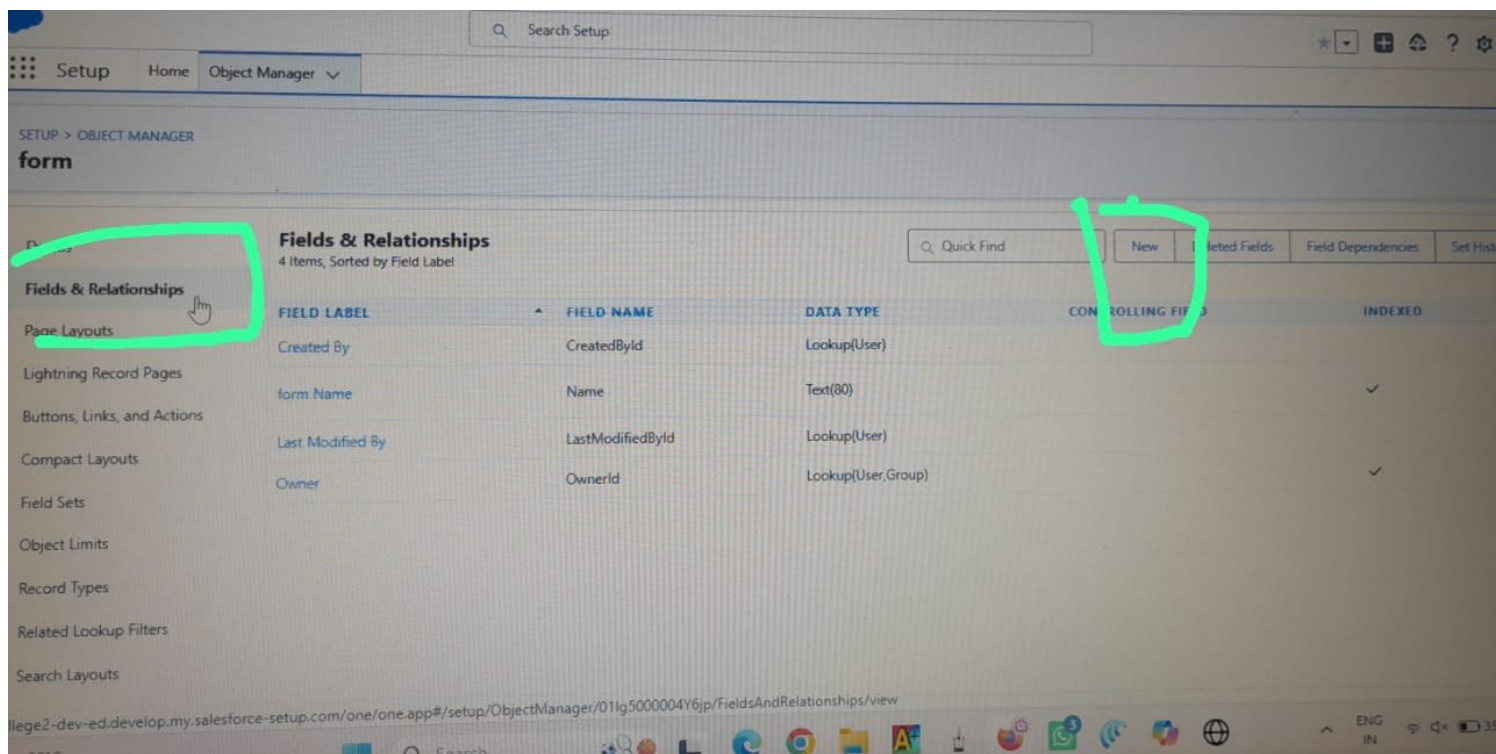
Milestone 5 - Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

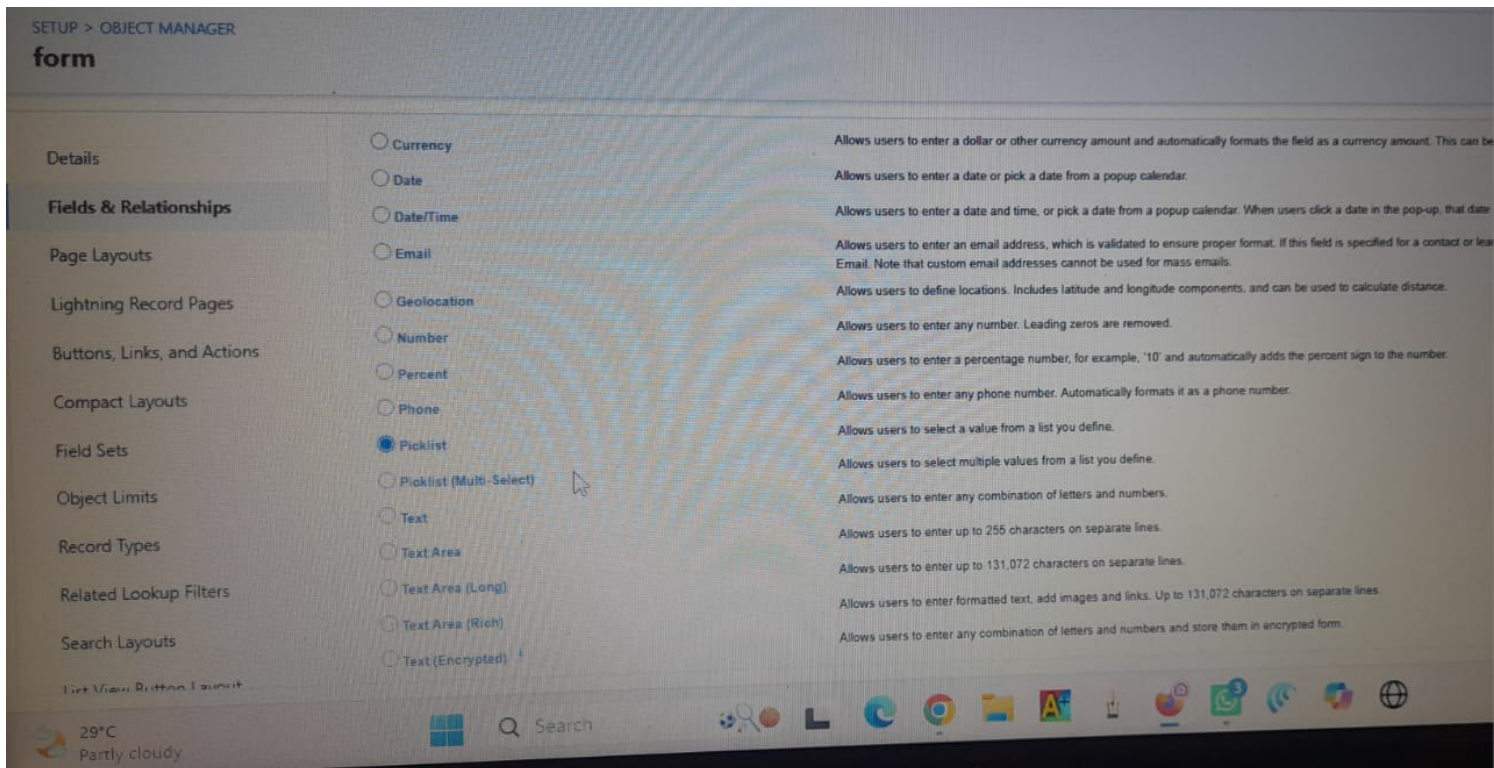
Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



5. Choose Data type as a picklist, Click next



6. Enter Field Label (Farm Type),and Enter Values as (Crop,Livestock,Mixed)
7. Next,Next Click Save.

SETUP > OBJECT MANAGER

form

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Buttons Layout

29°C Partly cloudy

Field Label: form type

Values:

- ☐ Use global picklist value set
- ☒ Enter values, with each value separated by a new line

Crop
Livestock
Mixed

☐ Display values alphabetically in the picklist

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field Name: form_type

Description:

Help Text:

Search

ENG IN 88% 22:05:00

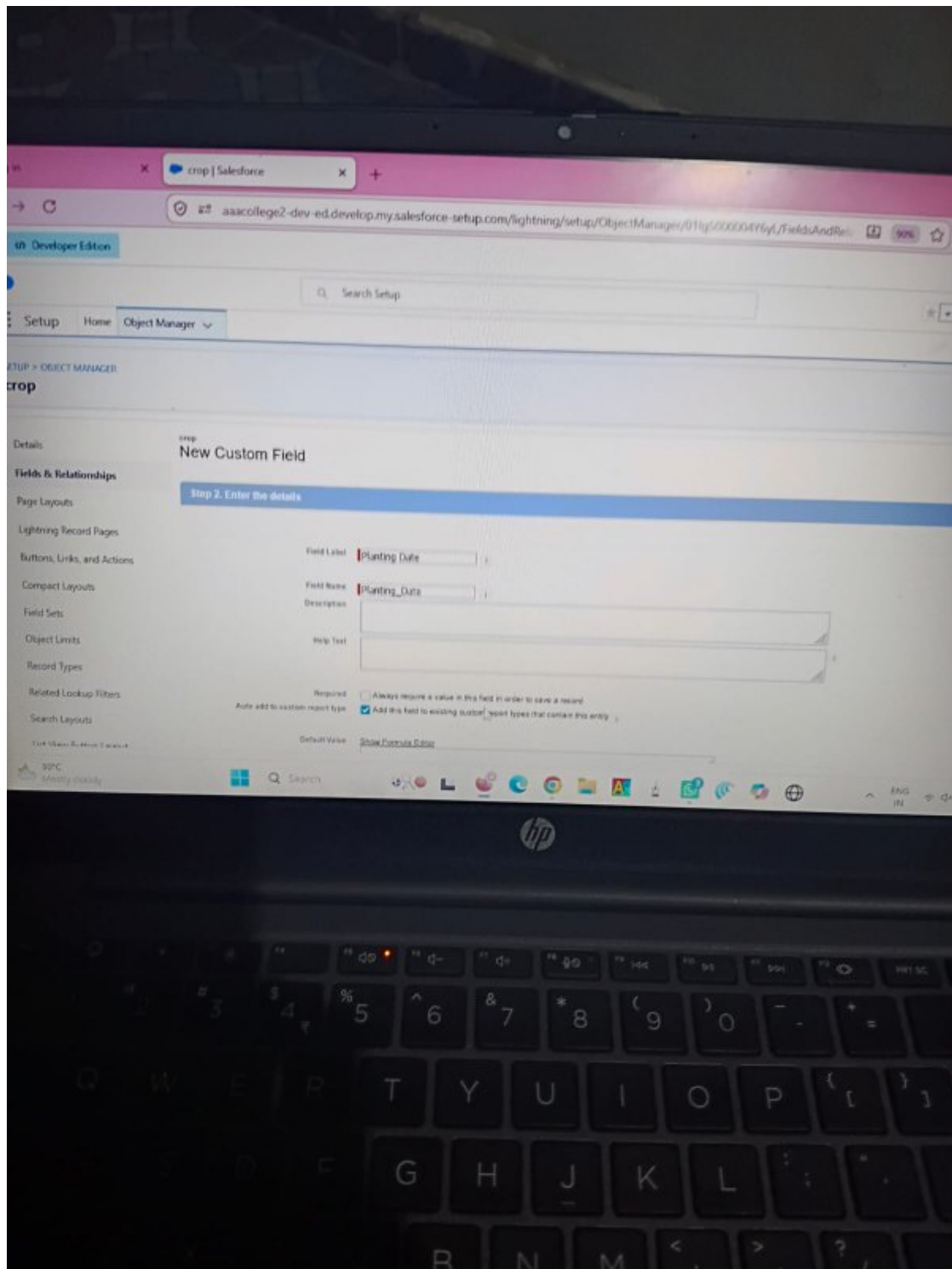
Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1. Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
2. Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
3. Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

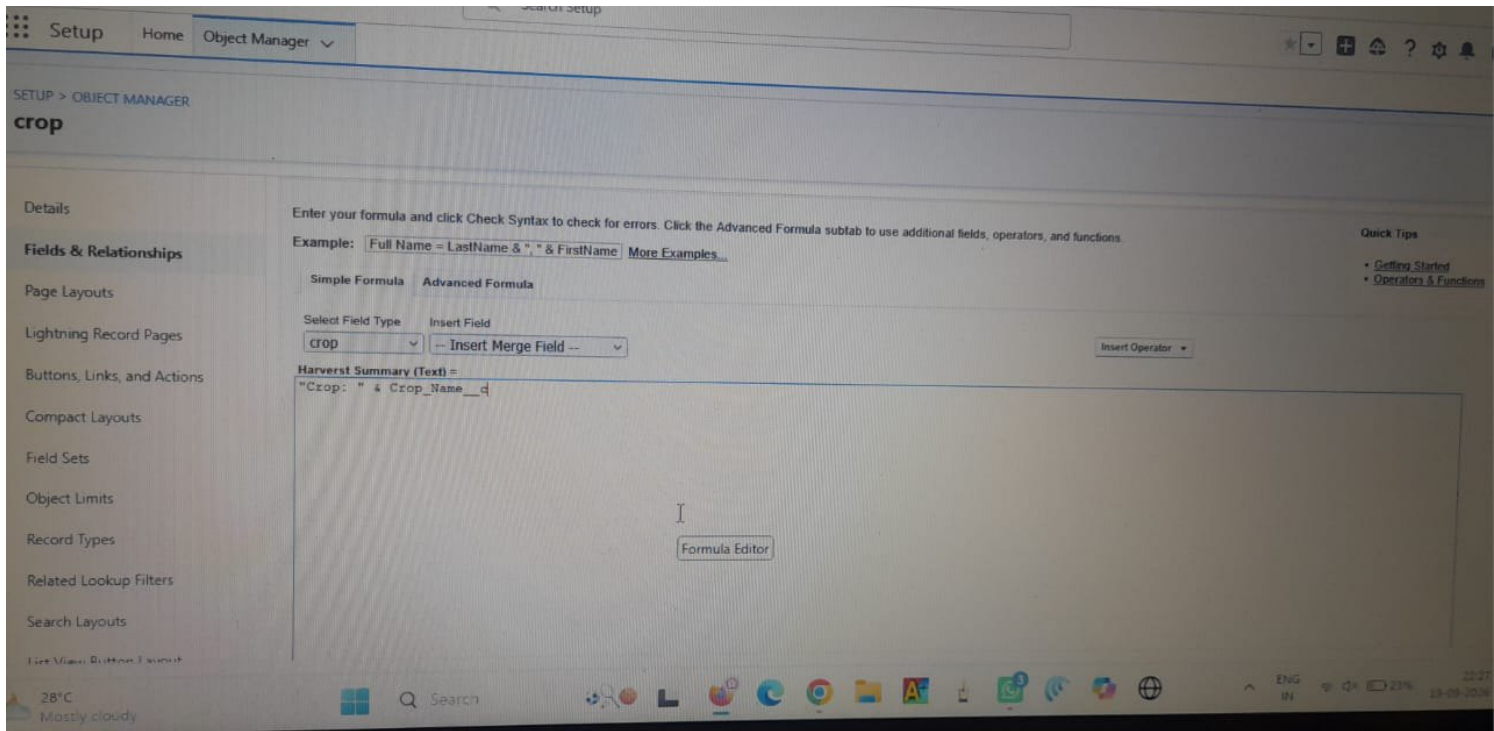
Activity 2:

Creation of fields for the Crop objects:

- 1.Choose Data type As a text, Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
3. Again we select data type as a date, Enter field labe (Harvest date),save
4. Next We Create Lookup relationship on crop object realted to Farm Object.
5. We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
6. Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
7. Now We create Formula Field For That Select data Type As a Formula,Click Next.



8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.
9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

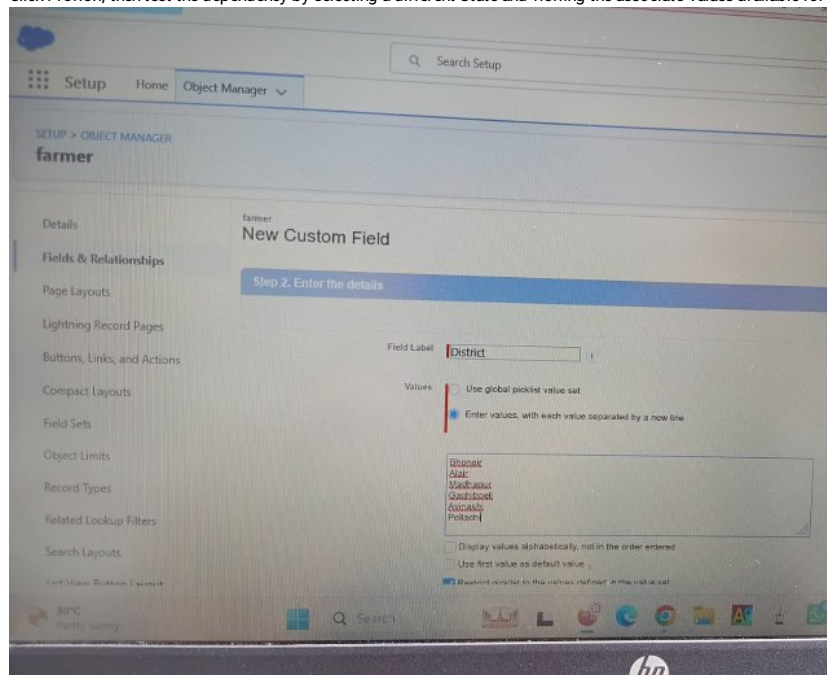
Creation of fields for the Farmer objects:

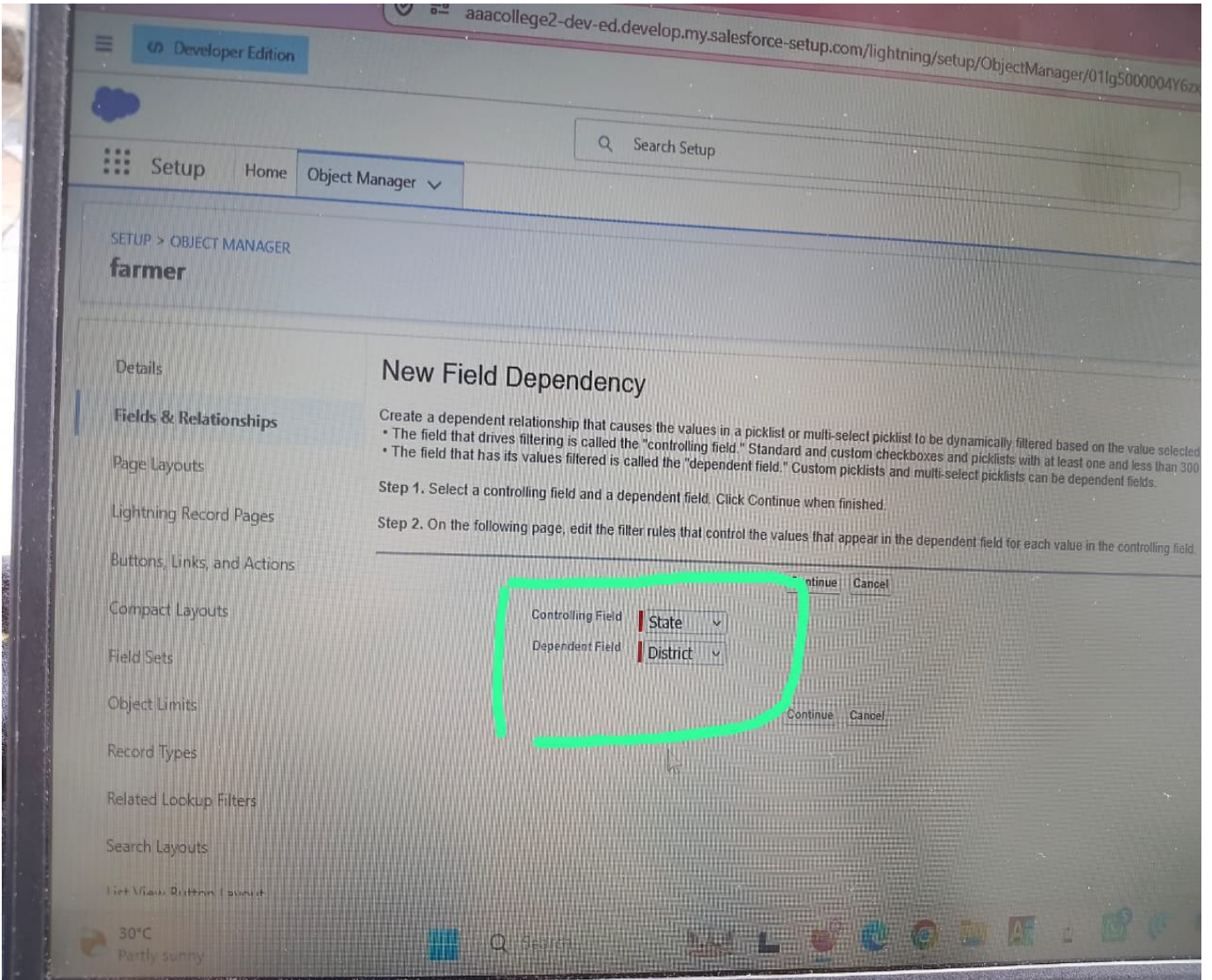
1. Select Data type Phone, Enter field label(Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label(State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again, select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency:

1. Now create Field Dependency for District field, Open Object Manager, Choose Farmer object, select Fields&Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field as District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

7.





- Included Value

Exclude Values

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All

Exclude Values

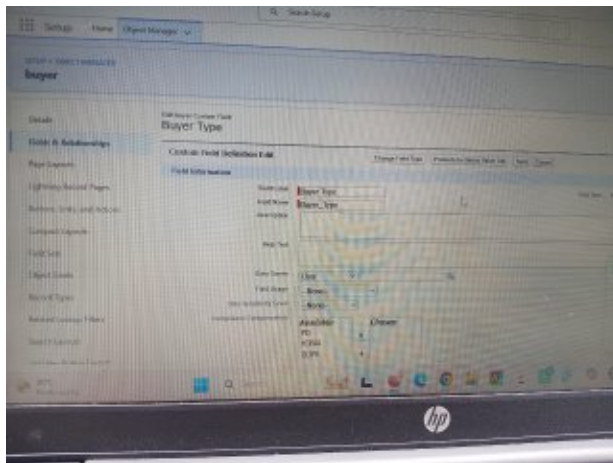
Preview

8. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone,Enter a Field Label(Contact),Save.
2. Select Data type as an Email,Enter Field Label(Email), Next save.
3. Select Data type Text area,Enter Field Label(Address),Save it.
4. Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.



Milestone 6- Profile:

What is a profile?

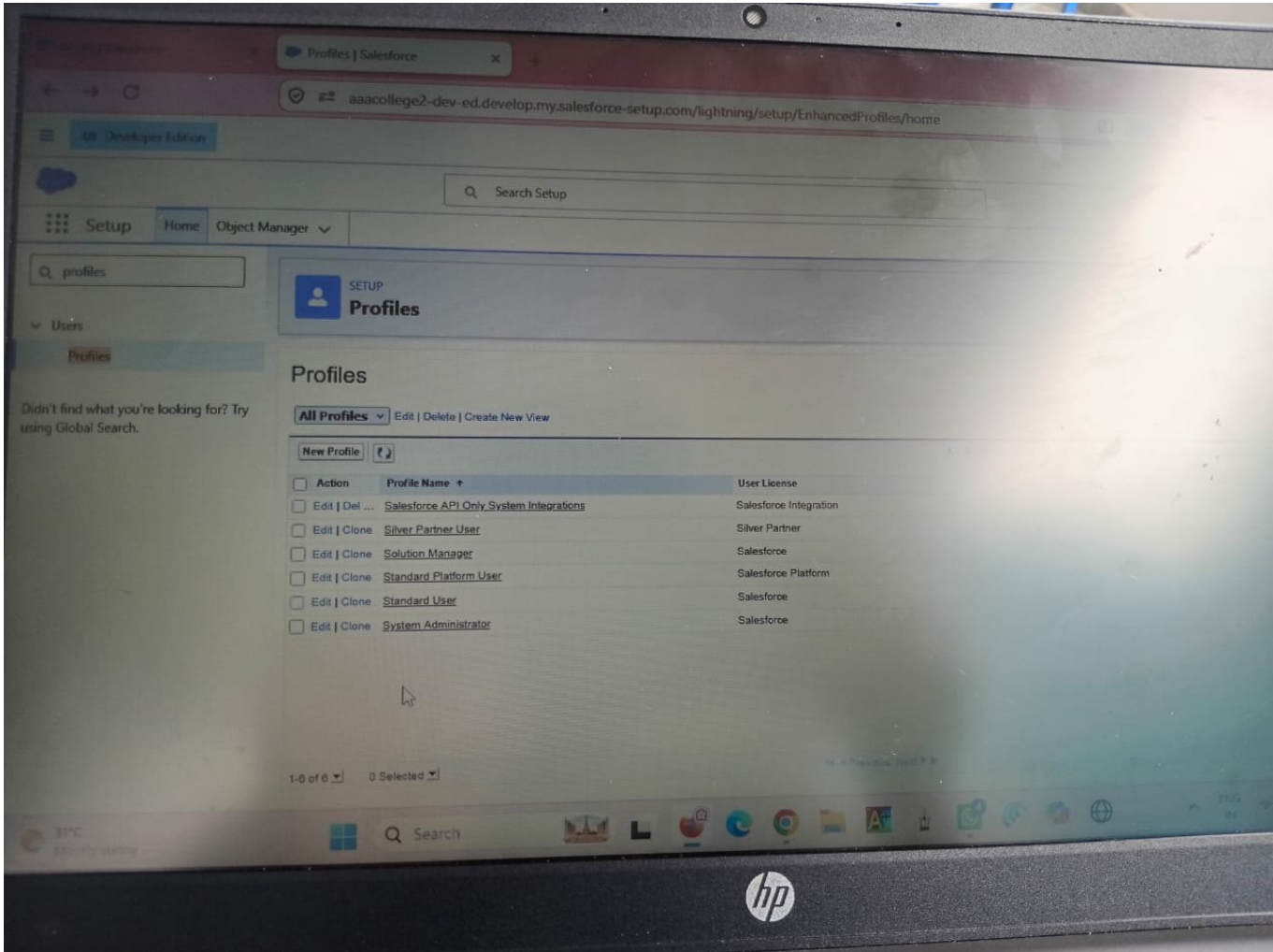
A profile in Salesforce defines the permissions and access levels assigned to a user. It controls which objects, fields, records, and system features a user can access. In the Farm Management System, profiles are used to provide appropriate access to Farm, Crop, Farmer, and Buyer objects. The System Administrator has complete access to

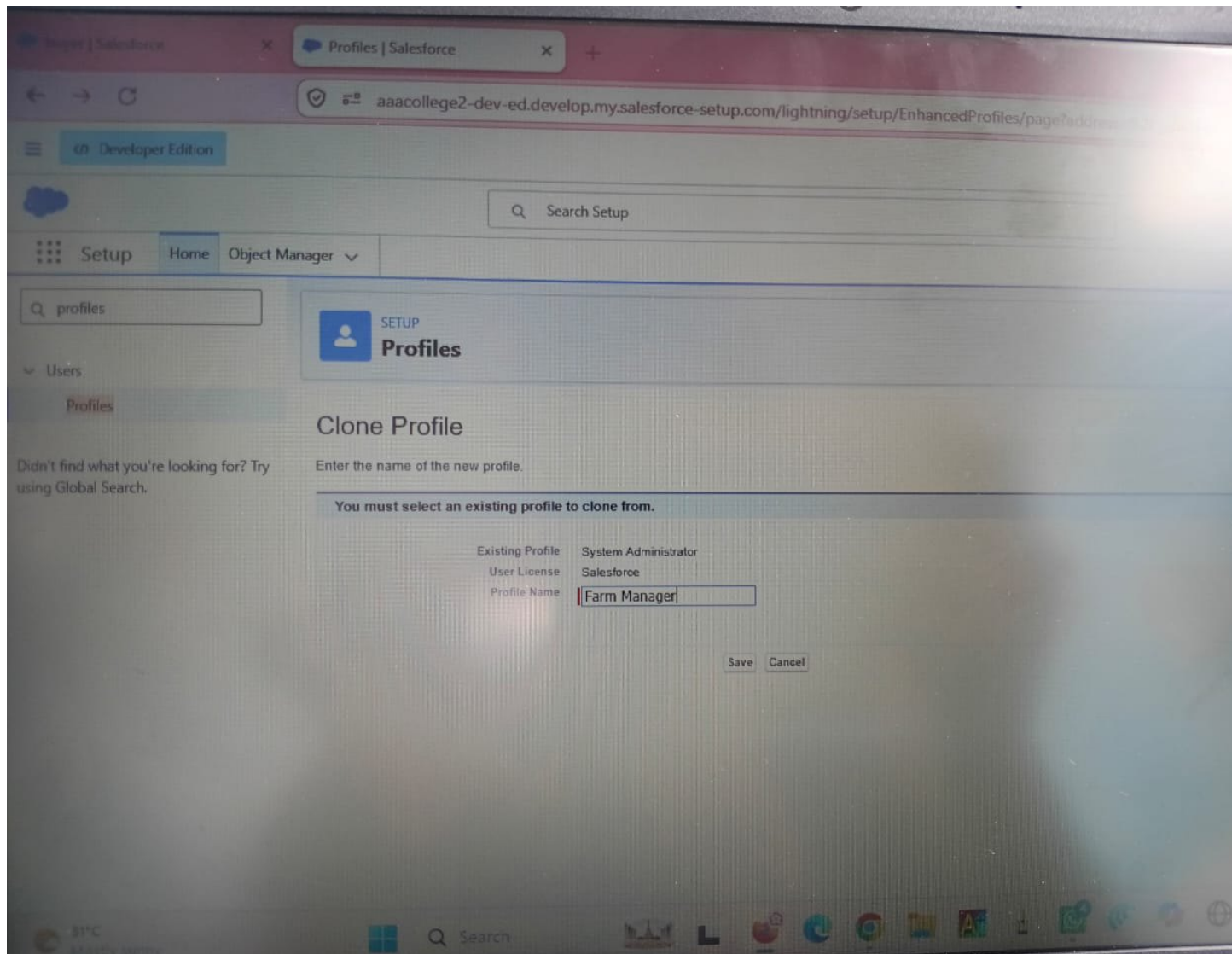
configure and manage the application, while other users are provided with permissions based on their responsibilities.

Activity 1:

Create a custom profile:

1. From setup, enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.





5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name "Farm Manager".

6. From setup, enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

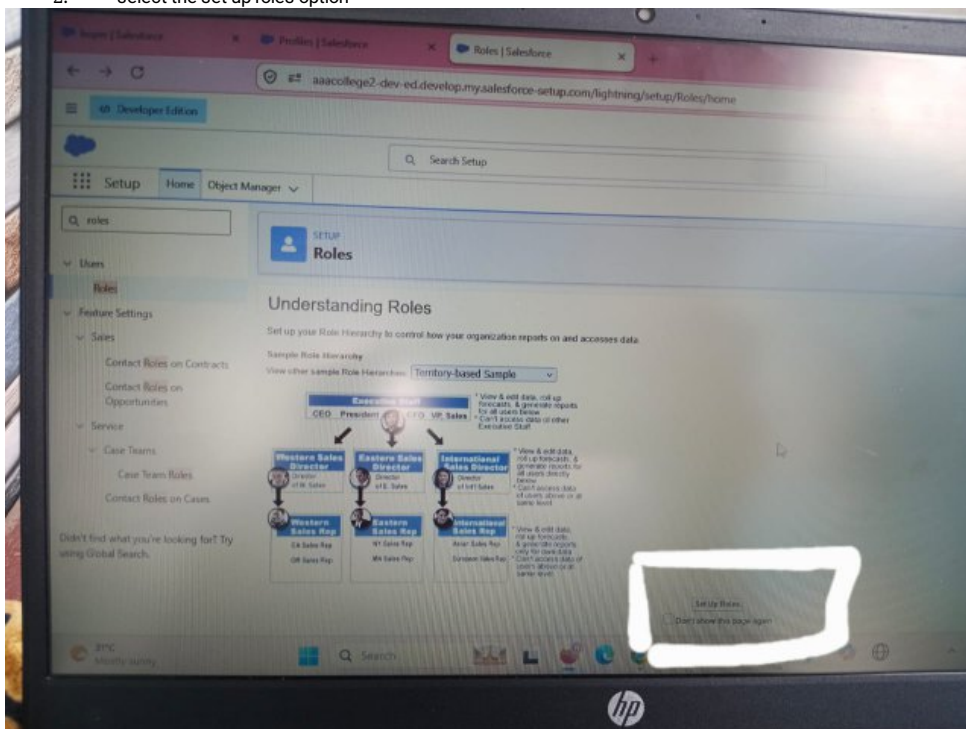
Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

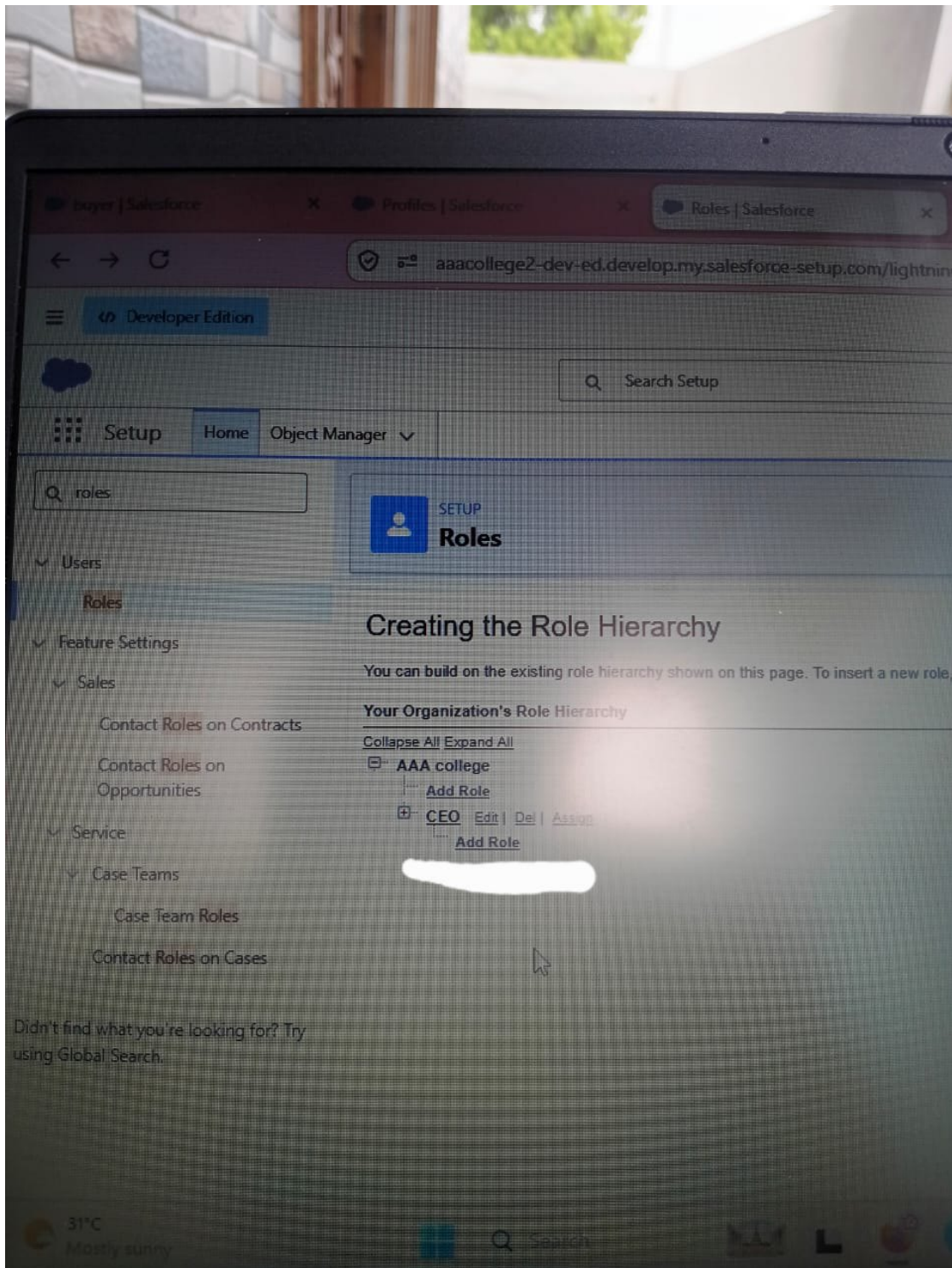
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3. Below the CEO click on add role and enter the label name as a "Farm Manager" and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

In Salesforce, a User is a person who has login access to the Salesforce organization. Each user is assigned a username, password, profile, and role based on their responsibilities.

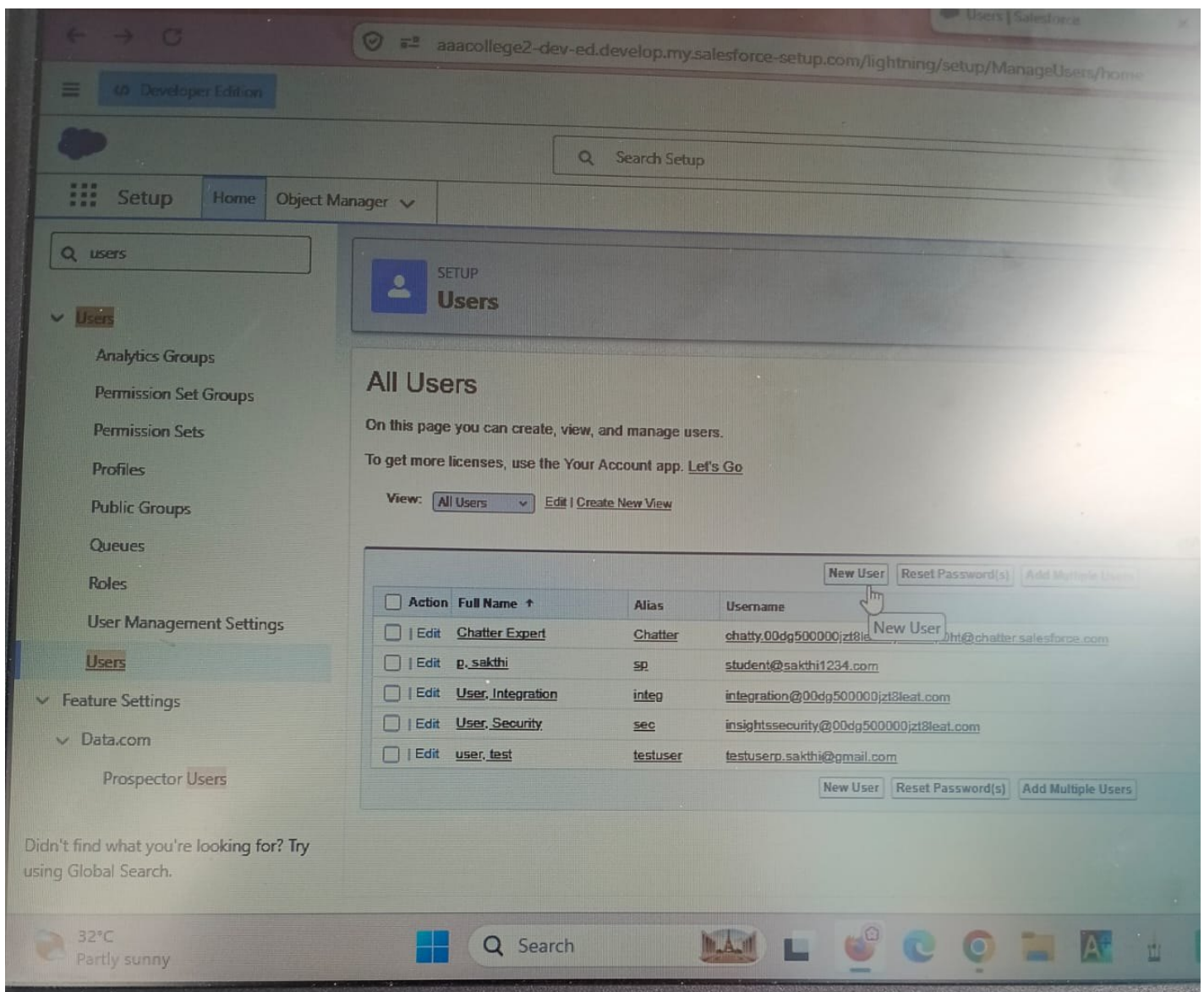
Users can access Salesforce features and perform actions according to the permissions assigned to them.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
 - 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
 - 3) Add the label of the sharing rule you want to make.
-
- 4) Select your rule type based on the criteria.
 - 5) Select the field can join immediately check field from the Crop object.
 - 6) Select the Crop as equal and value is True.
 - 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
 - 8) And in the section of select the level of access for the users give the access Read/Write.
 - 9) And save the rule.

aaacollege2-dev-ed.develop.my.salesforce-setup.com/lightning/setup/SecuritySharing/page?address=%2Fsetup%2F...

Developer Edition

Search Setup

Setup Home Object Manager

Security

Guest User Sharing Rule Access Report

Sharing Settings

SETUP

Sharing Settings

Step 1: Rule Name

Label: Crop Sharing with Farm M...

Rule Name: Crop Sharing with Farm M...

Description:

Step 2: Select your rule type

Rule Type: ☐ Based on record owner ☒ Based on criteria

Step 3: Select which records to be shared

Criteria	Field	Operator	Value	
	--None--	--None--		AND
	--None--	--None--		AND
	--None--	--None--		AND
	--None--	--None--		AND
	--None--	--None--		AND

Add Filter Logic...

Additional Options ☒ Include records owned by user who can't have an assigned role

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.
Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows a web browser window with the Salesforce interface. The browser's address bar displays the URL: `aaacollege2-dev-ed.develop.lightning.force.com/lightning/o/farmer_c/new?count=1&nooverride=1&useRecordTypeCheck`. The page title is "New farmer | Salesforce".

The main content area is a "New farmer" form. It has a title bar "New farmer" and a close button (X). Below the title bar is a section titled "Information" with a legend indicating that an asterisk (*) denotes required information.

The form contains the following fields:

- * farmer Name**: A text input field.
- Contact Number**: A text input field.
- State**: A dropdown menu currently showing "--None--". Below it is a link that says "View all dependencies".
- District**: A dropdown menu currently showing "--None--". Below it is a link that says "View all dependencies".

On the right side of the form, there is an "Owner" section showing a user icon and the name "sakthi p".

At the bottom right of the form, there are three buttons: "Cancel", "Save & New", and "Save".

The background of the browser window shows other Salesforce tabs and a sidebar with the "App Launcher" and "Farm Management" app.

Milestone 11-Reports:

What are Reports?

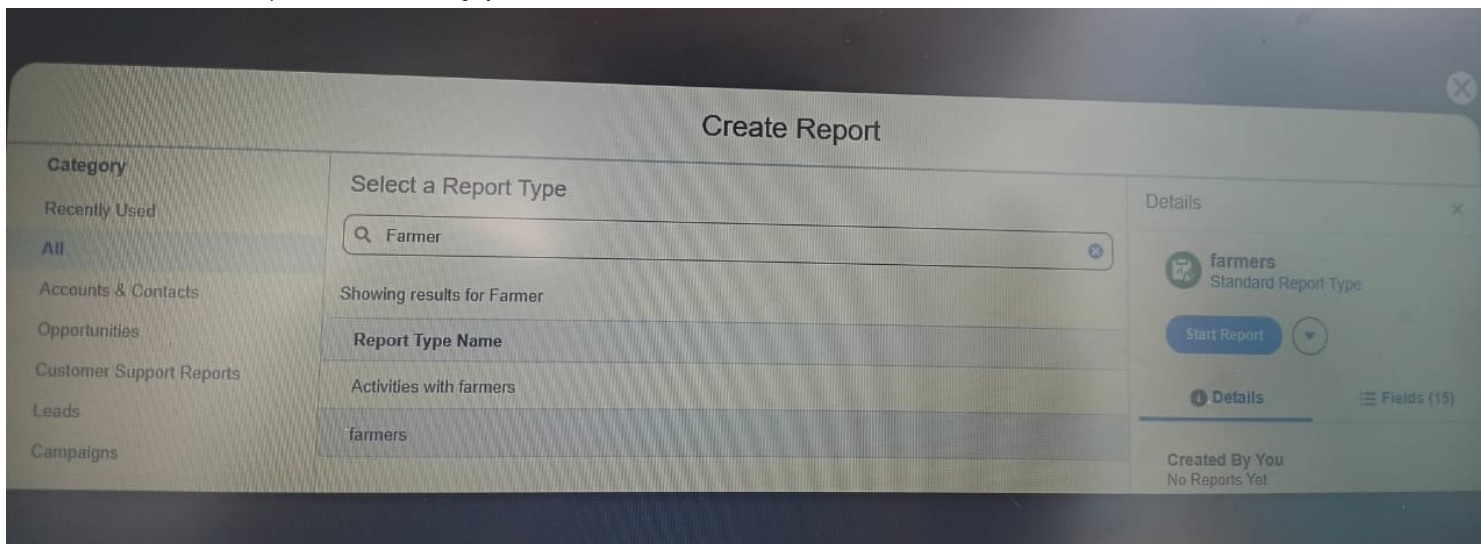
In Salesforce, a Report is a collection of records that matches the criteria defined by the user. Reports display data in rows and columns and can be filtered, grouped, and presented using charts. Every report is stored in a folder. Folders can be public, hidden, or shared, and access can be set as read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group, Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot shows the Salesforce Developer Edition interface. The browser address bar displays the URL: `aaacollege2-dev-ed.develop.lightning.force.com/one/one.app#eyJjb21wb251bnREZWYiOiJyZXBvcnRzOnJlcG9yd`. The top navigation bar includes a search bar and a menu with options: Home, Opportunities, Leads, Tasks, Files, Accounts, Contacts, Campaigns, and Dashboards. The left sidebar shows the 'Sales' app with a 'REPORT' dropdown menu. The 'crop Report' is selected, and the 'crops' group is active. The main content area displays a table with the following data:

crop: crop Name	Crop Name	Planting Date	Harverst Date	Actual Yield
1	paddy	20/09/2026	20/09/2026	₹50,000
2				₹50,000

Below the table, there is a 'Columns' section with a search bar and a list of fields: 'crop: crop Name', 'Crop Name', 'Planting Date', 'Harverst Date', and 'Actual Yield'. Each field has a corresponding 'X' icon to toggle its visibility. The bottom of the screen shows a Windows taskbar with various application icons and a search bar.

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.

crop Report

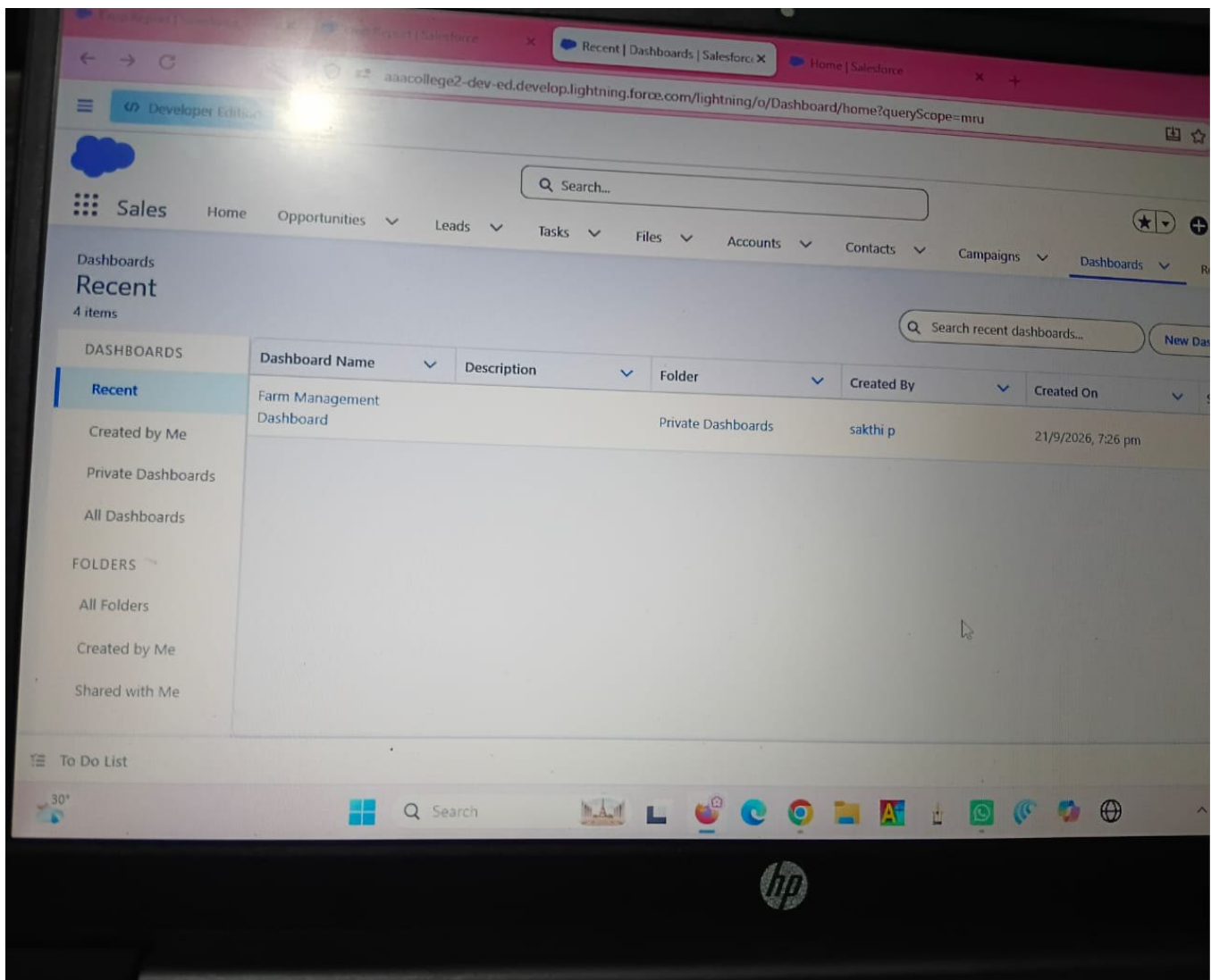
crop: crop Nam...	Crop N...	Planting ...	Harverst ...	Actual Y...
paddy	paddy	20/09/2026	20/09/2026	₹50k

View Report (crop Report)

Milestone-13: View Report and Dashboard:

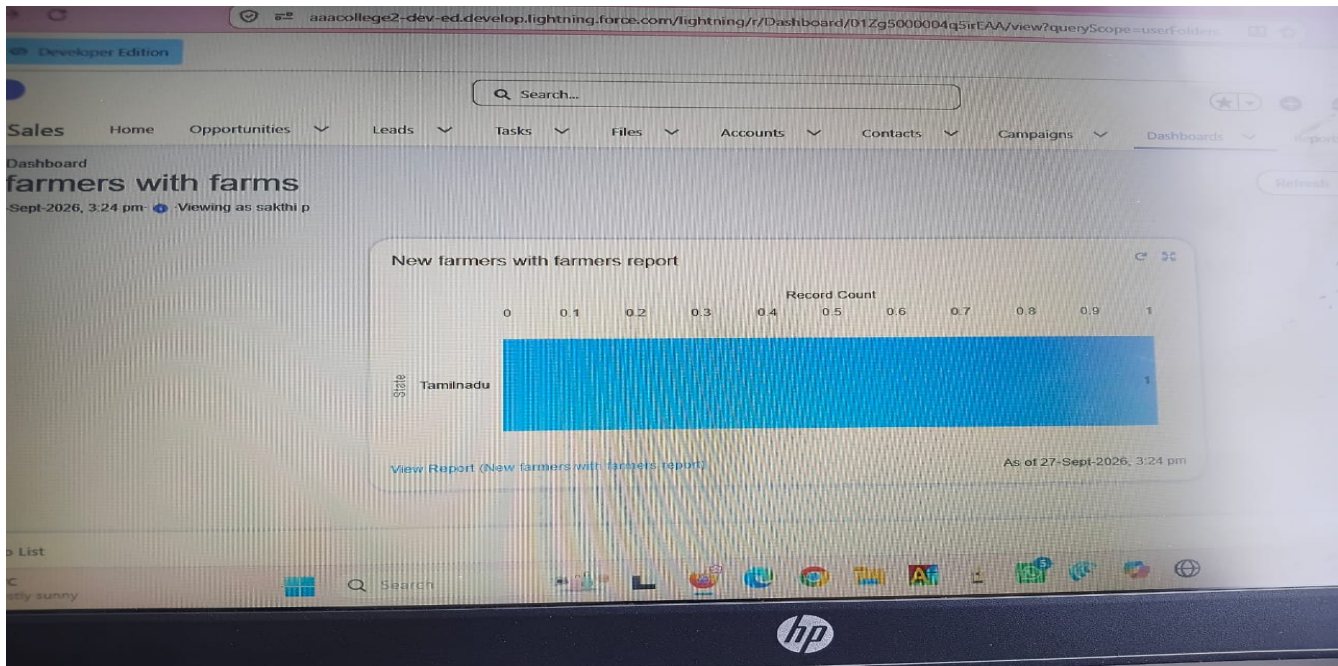
Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

The Functional and Performance Testing phase was conducted to verify that the Salesforce-Based Farm Management System works according to the specified requirements. Functional testing was performed to check the correctness of each module, while performance testing was carried out to ensure smooth system response during regular user operations.

Functional Testing

Creation of Farm, Crop, Farmer, and Buyer records.
Verification of required fields and data validation rules.
Testing of relationships between custom objects.
Checking user profiles, roles, and access permissions.
Verification of sharing rules and data visibility.
Testing of Reports and Dashboards.
Checking navigation through the Farm Management System App.

Performance Testing

Performance testing was conducted to ensure that:
Records load properly without unnecessary delays.
Users can access different modules smoothly.
Reports and dashboards display data correctly.
The system responds properly during multiple user operations.
Data remains stable and consistent during regular usage.

8. ADVANTAGES & DISADVANTAGES

Advantages

1. Requires an internet connection to access the system.
2. Users need basic knowledge of Salesforce.
3. Some advanced features may require paid Salesforce licenses.
4. Customization and configuration may require technical skills.
5. Offline access is limited

Disadvantages

1. The Functional and Performance Testing phase was conducted to verify that the Salesforce-
2. Based Farm Management System works according to the specified requirements.
3. Functional testing was performed to check the correctness of each module, while performance testing was carried out to ensure smooth system response during regular user operations.
4. Functional Testing

9. Conclusion:

The Salesforce-Based Farm Management System successfully digitizes and simplifies agricultural data management. The system helps manage Farm, Crop, Farmer, and Buyer records using Salesforce custom objects.

It improves data accuracy through validation rules, provides secure access using profiles and sharing rules, and supports better decision-making through Reports and Dashboards. Overall, the system reduces manual work and provides an organized platform for managing farm-related information.

10.FUTURE SCOPE

11. The future enhancements of the Farm Management System may include:
12. Mobile Application Integration – Access farm records through mobile devices.
13. AI-Based Crop Prediction – Predict suitable crops based on soil and environmental conditions.
14. Weather Data Integration – Display real-time weather information for farming activities.
15. IoT Sensor Connectivity – Monitor soil moisture, temperature, and other farm conditions.
16. Government Scheme Integration – Provide information about agricultural subsidies and government schemes.
17. Market Price Forecasting – Display current and predicted crop market prices.
18. Automated Notifications – Send reminders for crop activities, harvesting, and farm maintenance.
19. Advanced Analytics – Provide detailed insights through advanced reports and dashboards.

20.APPENDIX

Source Code

Salesforce configuration, custom objects, fields, relationships, validation rules, profiles, roles, sharing rules, reports, dashboards, and application settings.

Dataset

Farm, Crop, Farmer, and Buyer records created and maintained in the Salesforce platform.

Tools Used

Salesforce Developer Org
Salesforce Lightning Experience
Object Manager
Reports and Dashboards
Salesforce Flow
Validation Rules

Project Platform

Salesforce Cloud Platform

Thank You